

## User Task Description

**[User]** Simulate 5 residential buildings, use SAC algorithm to minimize electricity cost, then test and plot the voltage on the IEEE 33-bus system...

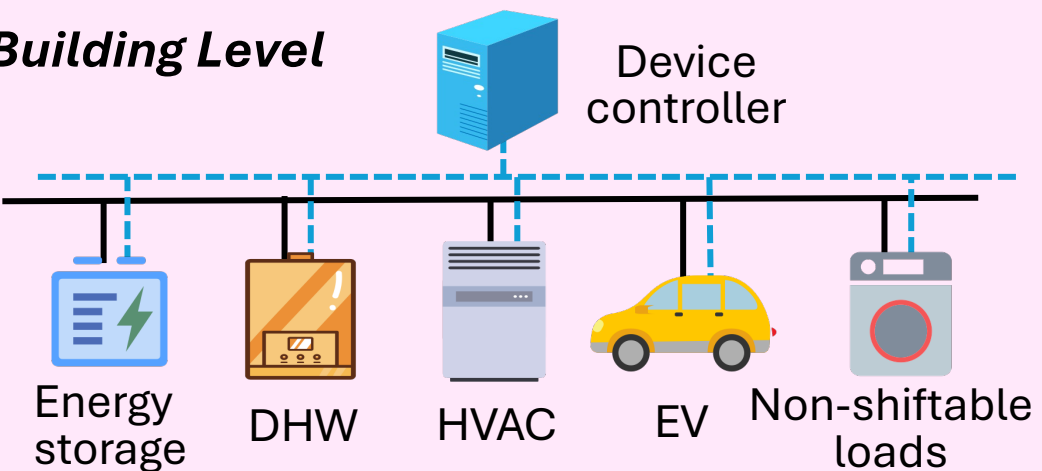
## Agentic Retrieval

**[Agent]** Searching **[codebase]**...  
**[Agent]** Output:  
{  
  "Stage1": [  
    "Function\_Name1", ...  
  ], ...  
}

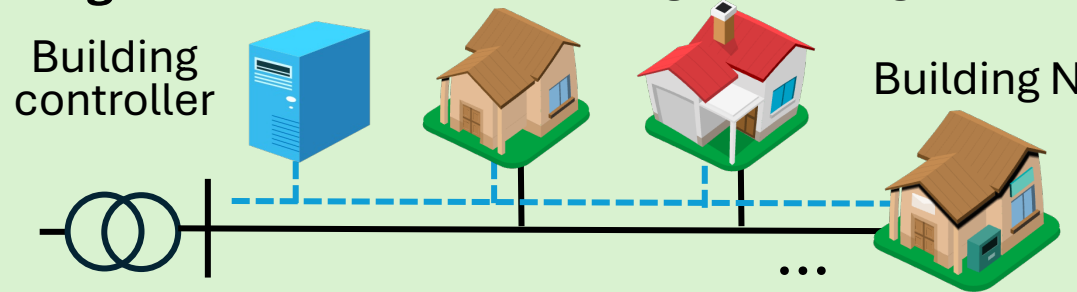
## Simulation Code Generation

**[SOCIA]** Output:  
Import...  
  
def Controller\_Configuration (...):  
  agent = RL\_Controller (...  
  return ...  
...

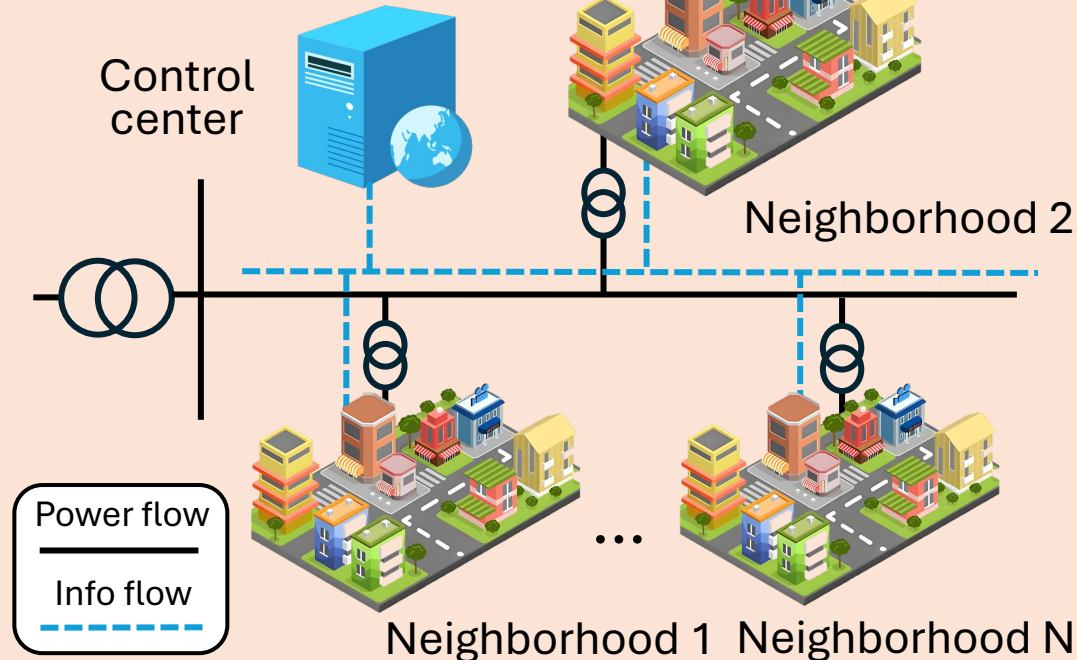
### Building Level



### Neighborhood Level



### Grid Level



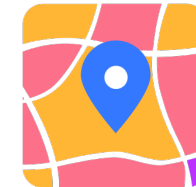
## Configuration Layer

### EnergyPlus-based Building-level Simulation

Building Configuration



Regional & Climatic Selection



Building Data Generation



## Control Layer

### CityLearn-based Neighborhood-level Control Simulation

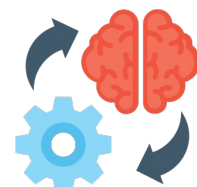
Environment Configuration



Algorithm Selection



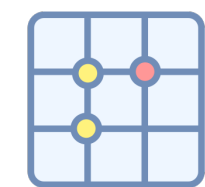
Model Training



## Evaluation Layer

### Pandapower-based Grid-level Power System Simulation

Grid Configuration



Power Flow Calculation



Grid Safety Analysis



## Simulation Output

### Result Management & Visualization

Simulation Metric Output



Figure Generation



Simulation Result Storage

